

atandra bharati

deep learning research engineer · 12 from-scratch PyTorch projects across LLMs, latent diffusion, multimodal, and agentic ML systems

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kolkata, india · remote-friendly · open to global remote

summary

self-taught deep learning research engineer with 12 from-scratch PyTorch projects across LLMs, latent diffusion, multimodal AI, and an agentic research platform. focused on GPU memory engineering, single-GPU feasibility, faithful paper reproduction, and agentic orchestration of ML research. headline results: 78% peak-memory cut on LLaMA-3 pretraining (92 GB to 20 GB), 0.0947 training loss on Stable Diffusion 1.x from scratch (860M UNet, 2x RTX 5090), and a 15-phase multi-agent ML research platform with 878 passing tests.

experience

ml engineering portfolio

nov 2022 – present · kolkata, india

self-directed · full-time · github.com/atandra2000
designed, implemented, and trained 12 end-to-end production-grade deep learning systems in raw PyTorch – no high-level wrappers, no pretrained shortcuts where the model itself is the subject. every project engineered to fit on a single A100 80GB, 2x RTX 5090, or smaller via BF16, Flash-Attn 2, gradient checkpointing, torch.compile, and chunked cross-entropy. coverage: LLM pretraining (DeepSeek-V3 reproduction, LLaMA-3, FusionLLM hybrid), latent diffusion (SD 1.x from scratch on 1.3M+ images across 7 phases), multimodal (PaliGemma-style VLM on COCO with zero pretrained weights), generative vision (CycleGAN, beta-VAE, DCGAN), video understanding (HRNet pose + ST-GCN), and an agentic ML research platform.

selected projects

five flagship builds · 2022–2026

stable diffusion 1.x from scratch

latent diffusion · generative vision

trained a complete SD 1.x model from random init on 2x RTX 5090, releasing the 42-epoch checkpoint. 860M-param UNet across a 7-phase curriculum on 1.3M+ images (LAION to high-aesthetic LAION to DiffusionDB/JourneyDB to VGGFace2 to COCO to consolidation); best loss 0.0947 at epoch 16. fault-tolerant multi-day training via atomic .tmp to os.replace checkpointing, 4-stage VAE-latent pre-encoding pipeline producing 32 KB latents, Min-SNR loss weighting, GPU-resident EMA (0.9999), channels_last on Blackwell, DDP/NCCL with no_sync accumulation.

860M UNet · 7 phases · 1.3M+ images 2x RTX 5090 · 42 epochs · best loss 0.0947

llama-3-lite: 78% memory reduction on single A100

llm · memory engineering

built a 515M-param LLaMA-3-style transformer engineered to fit on a single A100 80GB, cutting peak training memory by 78% (92 GB to 20 GB). the cut came from stacking gradient checkpointing, chunked cross-entropy (logits 50 GB to 0.3 GB), and disk-backed token caching (RAM 112 GB to 1 MB) – enabling 2x batch-size headroom and roughly 33% throughput gain. gqa (8Q/4KV), RoPE (theta=500K), fused SwiGLU, RMSNorm, Flash-Attn 2, BF16, safetensors with full RNG-state restore.

515M params · 8.25B-token run A100 80GB · chunked CE · bf16 · FA2

autonomous ml research engineer

agentic platform · research infra

15-phase multi-agent platform that turns an arxiv paper into evaluated ML experiments end-to-end – paper analysis, repo analysis, experiment planning, code patches, training runs, statistical evaluation, autonomous looping, end-to-end research reports. provider-agnostic LLM layer (qwen3-coder, glm-5.2, ollama cloud) with per-agent model routing, vector plus graph memory, pydantic v2 schemas, self-repair. 878 passing tests across the agent and tool surface.

23 agents · 61 tools · 186 models · 878 tests qwen3-coder · glm-5.2 · ollama cloud · pydantic v2

deepseek-v3-lite: faithful V3 reproduction

llm · frontier architecture

faithful from-scratch reimplementation of DeepSeek-V3 at 422M params – MLA, aux-loss-free MoE, MTP, end-to-end. MLA with KV low-rank compression (kv_lora_rank=192), the absorption trick, decoupled RoPE; DeepSeekMoE (20 routed top-4 plus 2 shared) with aux-loss-free biased-sigmoid balancing; MTP with true absorption at inference. 8.4B-token chinchilla recipe on a single A100 80GB via SDPA Flash-Attn 2, muP LR scaling, torch.compile(max-autotune); speculative-decoding inference (up to 2x throughput via MTP draft head). author of the 643-line MLA technical deep-dive.

422M params · mla · moe · mtp A100 80GB · sdpa · torch.compile · muP · 8.4B-token recipe

fusionllm: hybrid mla + gdn + moe + mtp

llm · hybrid architecture

24-layer single-GPU pre-training framework fusing four frontier mechanisms in one model: MLA, Gated Delta Net (linear-complexity attention via chunked delta-rule), DeepSeek-style MoE, and Multi-Token Prediction. 415.6M active / 868.6M stored params, 64K vocab, seq_len 4096. dual-optimizer strategy (NorMuon for 2D matrices, CautiousAdamW for embeddings and norms), WSD schedule, muP init, logit softcap +−15.0, FP32-precise GDN recurrence. 6-stage resumable data pipeline, 55 unit tests.

415.6M active / 868.6M stored

NorMuon + CautiousAdamW · WSD + muP · 8.31B-token recipe

additional projects

seven more · each with verified repo and build log

- vision-language model (PaliGemma-inspired) · siglip vision encoder + linear projector + gemma-style gqa decoder · trained end-to-end on coco 2014 captions with zero pre-trained weights · [P100](#)
- face aging cyclegan · per-layer adain conditioning on a 101-class age embedding, 3-scale patchgan + mean-residue age loss, lsgan + vgg-19 perceptual + l1 identity · 31/50 epochs on imdb-wiki (500k faces) · [RTX 6000 Ada](#)
- face generation beta-vae · 128x128 celeba, bilinear-upsample decoder (no convtranspose), linear kl annealing 0 to 1 over 30 epochs · recon mse 0.0152 · [P100](#)
- dcgan face generation · faithful 2015 paper reproduction (radford et al.), 50 epochs on 202k celeba, d loss converges to ln 2 ~ 0.693 (theoretical nash equilibrium) · [2x T4 Kaggle](#)
- neural machine translation (en to it) · encoder-decoder transformer (6+6, d_model=512), custom sentencepiece bpe, cross-attention visualization · loss 6.17 to 2.28 over 20 epochs on opus_books · [P100](#)
- action recognition (st-gcn) · real-time skeleton-based action recognition with from-scratch hrnet-like pose estimation + two-stream ctr-gcn · 120-class NTU RGB+D 120 pipeline ready, real-time inference targeting 30 FPS on RTX 3090 · [RTX 3090](#)

technical skills

built up across twelve from-scratch projects · every keyword has shipped code behind it

llm	LLaMA-3, DeepSeek-V2/V3, MLA, GQA, MQA, MoE, MTP, Gated Delta Net, Chinchilla scaling, muP, WSD, NorMuon, AdamW, speculative decoding
generative vision	Stable Diffusion 1.x, latent diffusion, UNet, DDPM, DDIM, Min-SNR, classifier-free guidance, EMA, VAE, GAN, DCGAN, PatchGAN, beta-VAE, CycleGAN, AdaIN, spectral normalization, VGG and LPIPS perceptual loss
multimodal and video	ViT, SigLIP, PaliGemma-style fusion, HRNet pose estimation, ST-GCN, CTR-GCN, graph convolution, NTU RGB+D 120
transformer components	RoPE, GQA, SwiGLU, GeGLU, RMSNorm, multi-head attention, MLA, MQA, MTP, cross-attention
training and infra	BF16, FP16, FP8, mixed precision, DDP, FSDP, NCCL, torch.compile, Flash-Attn 2, SDPA, gradient checkpointing, chunked cross-entropy, disk-backed token cache, atomic checkpointing, resumable pipelines
agentic ml	multi-agent orchestration, provider-agnostic LLM routing, ollama cloud, vector and graph memory, pydantic v2, pytest, ruff
core	Python 3.12, PyTorch 2.x, HuggingFace, diffusers, safetensors, sentencepiece, NumPy, W&B, Comet ML, img2dataset, Git, GitHub
hardware validated	A100 80GB, RTX 5090 (Blackwell), RTX 6000 Ada, RTX 3090, P100, T4 (2x), RunPod, Kaggle

writing

multi-head latent attention: a technical deep-dive

643-line reference on kv-cache math, low-rank compression algebra, the absorption-trick derivation, and decoupled rope mechanics in deepseek-v2/v3. read on github.

education

b.tech, civil engineering

2020 – 2024

heritage institute of technology, kolkata · [heritageit.edu](#)

civil engineering for the math, structures, and optimization; deep learning self-taught in parallel, from raw PyTorch up to frontier architectures.